THE INDIAN INTERNATIONAL SCHOOL,DSO

Worksheet – Chapter 6: Application of Derivatives (2026-2027)

Grade: 12 Subject: MATHEMATICS

Name: _____

Date: _____

Instructions:

- This worksheet has 20 questions arranged in three levels of increasing difficulty.
- Level 1: Q 1 – Q 7 (MCQs, Assertion–Reason and short answer questions)
- Level 2: Q 8 – Q 14 (short answer questions)
- Level 3: Q 15 – Q 20 (long answer and case-based questions)
- Marks for each question are given against it. All questions are compulsory.
- The answer key is given at the end of the worksheet.

Q No.	Question	Marks
	LEVEL 1 **Q 1 – Q 7**	**8**
1.	A cylindrical tank is being filled with sand at the rate of 314 m^3/h. If the radius of the tank is 10 m, then the height of sand in the tank increases at the rate of: (Take $\pi = 3.14$) (A) 1.1 m/h (B) 1 m/h (C) π m/h (D) $\frac{\pi}{2}$ m/h	1
2.	The function $f(x) = x^3 - 3x^2 + 3x - 100$ is: (A) increasing on $\mathbb{R}$ (B) decreasing on $\mathbb{R}$ (C) increasing on $(-\infty, 1)$ only (D) decreasing on $(1, \infty)$	1
3.	The interval in which $f(x) = x^2 - 4x + 6$ is strictly decreasing is: (A) $(-\infty, 2)$ (B) $(2, \infty)$ (C) $(-\infty, \infty)$ (D) $(-2,2)$	1
4.	The maximum value of $\sin x \cos x$ is: (A) 1 (B) $\frac{1}{2}$ (C) 2 (D) $\frac{1}{\sqrt{2}}$	1
5.	The local minimum value of $f(x) = x + \frac{1}{x}$, $x > 0$, is: (A) 1 (B) -2 (C) 2 (D) 0	1

Q No.	Question	Marks
6.	**Assertion (A):** $f(x) = \sin x$ is strictly increasing in $\left(0, \frac{\pi}{2}\right)$. **Reason (R):** If $f'(x) > 0$ for every x in an open interval, then f is strictly increasing in that interval. (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.	1
7.	The edge of a cube is increasing at the rate of 3 cm/s. How fast is the volume increasing when the edge is 10 cm long?	2
	LEVEL 2 **Q 8 – Q 14**	**19**
8.	Find the intervals in which $f(x) = 2x^3 - 3x^2 - 36x + 7$ is (a) strictly increasing (b) strictly decreasing.	3
9.	Air is pumped into a spherical balloon at the rate of 900 cm³/s. Find the rate at which the radius is increasing when the radius is 15 cm.	2
10.	Find the local maximum and local minimum values of $f(x) = x^3 - 6x^2 + 9x + 15$.	3
11.	Find the intervals in which the function $f(x) = \frac{\log x}{x}$ is strictly increasing or strictly decreasing.	3
12.	A ladder 5 m long leans against a wall. The bottom of the ladder is pulled along the ground away from the wall at 2 cm/s. How fast is its height on the wall decreasing when the foot of the ladder is 4 m from the wall?	3
13.	Find the absolute maximum and absolute minimum values of $f(x) = \sin x + \cos x$ on $[0, \pi]$.	3
14.	Find two positive numbers whose sum is 15 and the sum of whose squares is minimum.	2
	LEVEL 3 **Q 15 – Q 20**	**25**
15.	Show that of all rectangles inscribed in a given fixed circle, the square has the maximum area.	5
16.	Show that the height of the cylinder of maximum volume that can be inscribed in a sphere of radius R is $\frac{2R}{\sqrt{3}}$. Also find the maximum volume.	5
17.	A window is in the form of a rectangle surmounted by a semicircular opening. The total perimeter of the window is 10 m. Find the dimensions of the window so as to admit maximum light through the whole opening.	5
18.	Find the intervals in which $f(x) = \sin x + \cos x$, $0 \leq x \leq 2\pi$, is strictly increasing or strictly decreasing.	3
19.	A man 2 m tall walks at a uniform speed of 5 km/h away from a lamp post 6 m high. Find the rate at which the length of his shadow increases.	3

Q No.	Question	Marks
20.	**Case Study:** A technical company is designing a rectangular solar panel installation on a roof using 300 metres of boundary material. The design includes a partition running parallel to one of the sides, dividing the area into two sections. Let the length of the side perpendicular to the partition be x metres and the side parallel to the partition be y metres. **(i)** Write the equation for the total boundary material used (boundary and partition) in terms of x and y. **[1]** **(ii)** Write the area of the solar panel as a function of x. **[1]** **(iii)** (a) Find the critical point of the area function and use the second derivative test to find the maximum area. **OR** (b) Using the first derivative test, find the maximum area the company can enclose with the 300 m of boundary material. **[2]**	4

***** End of Worksheet *****

ANSWER KEY

Chapter 6: Application of Derivatives

Q No.	Answer / Key Steps
	LEVEL 1
1.	(B) $\frac{dh}{dt} = \frac{314}{\pi(10)^2} = 1$ m/h
2.	(A) $f'(x) = 3(x-1)^2 \geq 0$
3.	(A) $f'(x) = 2x - 4 < 0$ for $x < 2$
4.	(B) $\frac{1}{2}\sin 2x \leq \frac{1}{2}$
5.	(C) at $x = 1$, value 2
6.	(A) Both true; R explains A.
7.	$\frac{dV}{dt} = 3x^2\frac{dx}{dt} = 3(100)(3) = 900$ cm³/s
	LEVEL 2
8.	$f'(x) = 6(x-3)(x+2)$. Increasing on $(-\infty,-2) \cup (3,\infty)$; decreasing on $(-2,3)$.
9.	$\frac{dr}{dt} = \frac{900}{4\pi(15)^2} = \frac{1}{\pi}$ cm/s
10.	$f'(x) = 3(x-1)(x-3)$. Local max at $x = 1$: 19; local min at $x = 3$: 15.
11.	$f'(x) = \frac{1-\log x}{x^2}$; strictly increasing on $(0,e)$, strictly decreasing on (e,∞).
12.	$x^2 + y^2 = 25$; at $x = 4$, $y = 3$: $\frac{dy}{dt} = -\frac{4\times 2}{3} = -\frac{8}{3}$ cm/s. Height decreases at $\frac{8}{3}$ cm/s.
13.	Critical point $x = \frac{\pi}{4}$. $f(0) = 1$, $f\left(\frac{\pi}{4}\right) = \sqrt{2}$, $f(\pi) = -1$. Max $\sqrt{2}$, min -1.
14.	$S = x^2 + (15-x)^2$, $S' = 0 \Rightarrow x = 7.5$; numbers 7.5 and 7.5.
	LEVEL 3
15.	Sides $2r\cos\theta$, $2r\sin\theta$; $A = 2r^2\sin 2\theta$, maximum at $\theta = \frac{\pi}{4}$, i.e. length = breadth (square).
16.	$V = \pi h\left(R^2 - \frac{h^2}{4}\right)$, $V' = 0 \Rightarrow h = \frac{2R}{\sqrt{3}}$, $V'' < 0$. Max volume $= \frac{4\pi R^3}{3\sqrt{3}}$.
17.	$2r + 2h + \pi r = 10$, $A = 2rh + \frac{\pi r^2}{2}$. Max at $r = \frac{10}{\pi+4}$ m: width $\frac{20}{\pi+4}$ m, height of rectangle $\frac{10}{\pi+4}$ m.
18.	$f'(x) = \cos x - \sin x$. Increasing on $\left[0,\frac{\pi}{4}\right) \cup \left(\frac{5\pi}{4}, 2\pi\right]$; decreasing on $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$.
19.	Similar triangles: $\frac{s}{2} = \frac{x+s}{6} \Rightarrow s = \frac{x}{2}$; $\frac{ds}{dt} = \frac{1}{2}(5) = 2.5$ km/h.
20.	(i) $2x + 3y = 300$. (ii) $A(x) = x \cdot \frac{300-2x}{3} = 100x - \frac{2x^2}{3}$. (iii) $A'(x) = 100 - \frac{4x}{3} = 0 \Rightarrow x = 75$, $A'' = -\frac{4}{3} < 0$ (or A' changes sign from $+$ to $-$ at $x = 75$); $y = 50$; maximum area $= 3750$ m².